

Global Hackathon on Using AI for Digital Trade Regulatory Analysis

Hands on workshop and assignment feedback

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A

Bridge from theory to practice

Connecting legal identification techniques to your canvas

B

Assignment 1 review

Common patterns and mistakes across teams

C

Assignment 2 review

Some things to consider

D

Why is this output incorrect? (theoretical mistakes made by automation tool)

2 hands on exercises

E

Which output would you record in the RDTII database?

1 hands on exercise: let's think like lawyers!

A · Bridge from theory to practice

What the theory and hands on exercise covered

- ✓ Economy's legal system
- ✓ Source Hierarchy
- ✓ Source Validity (enforced law)
- ✓ Extracting and interpreting legal text

Where this is relevant on your canvas

Inputs ★

Legal sources and seed URLs

Outputs ★

Which source is recorded? Which source is the controlling evidence? Which source is currently active?

Evaluation ★

How to check if tool found the right output

Risks

When legal language tricks the tool

B · Assignment 1 review: Why is this output incorrect?

Singapore. Pillar 7 - Indicator 7.2: Lack of dedicated legal framework for cybersecurity

URL: <https://www.mas.gov.sg/regulation/notices-and-guidelines/cyber-hygiene-notices-2019> Title: MAS Notice on Cyber Hygiene (2019) Type: other Relevance Score: 0.85

Text: MAS introduced legally binding Notices on Cyber Hygiene in 2019, requiring all financial institutions to implement a set of baseline cybersecurity measures. For example, FIs must establish robust access controls (including multi-factor authentication), promptly apply security patches to systems, deploy malware protection, and maintain network security monitoring. These requirements are mandatory and enforceable across the financial sector.

Relevance Notes

This evidence shows a dedicated cybersecurity framework at the sectoral level. The financial sector regulator (MAS) has imposed specific cybersecurity obligations on financial institutions, demonstrating that beyond the general cybersecurity law, Singapore enforces cybersecurity requirements through sector-specific regulations.

Incorrect URL: The page could not be found so the URL is incorrect

Outdated law: When we manually looked up the law online, we found that the law is outdated: the notice was canceled since 01 July 2022 (<https://www.mas.gov.sg/regulation/notices/notice-1119>). Latest notice is: <https://www.mas.gov.sg/regulation/notices/notice-fsm-n16>

And these 2 reasons make the output incorrect, and a researcher would not record this MAS Notice. The researcher would record the latest Notice FSM-N16 Cyber Hygiene.

B · Assignment 1 review: Why is this output incorrect?

✓ What most teams got right

- Identified sectoral law is not controlling evidence when a horizontal law (Cybersecurity Act 2018) is present (in this example).
- Understood the correct legal question asked by the indicator
- Understood scoring criteria
- Understood that Notices are legal instruments too
- Extra note: Inversion of polarity

Please keep in mind that the indicators are designed to capture regulatory measures that can affect the ease of doing business for firms engaged in digital trade. As a general rule of thumb, a higher RDTII score indicates greater regulatory complexity, which may translate into a more costly business environment. Conversely, a lower RDTII score generally reflects a less complex/layered regulatory framework and a more enabling environment for digital trade.

We are not judging. Some laws while create compliance costs are necessary for public trust, market access and competition.

⚠ Common gaps

- Didn't notice that the regulation was outdated and should have been replaced with the new Notice that replaced it.
- Didn't notice that the URL was wrong
- Assumption that sectoral regulations are not relevant – They are relevant and recorded even if cross-cutting/ comprehensive framework is available.
- Assuming that regulations are lower than Acts in priority: Not really, any legal text published by government body is recorded. Whether it is controlling evidence or not is determined on a case-by-case basis. Example: Cybersecurity law (2018) is the controlling evidence for final indicator scoring but we will record MAS Notice FSM-N16 Cyber Hygiene because it does give information on sectoral cybersecurity measure.

C · Assignment 2 review: Tool Canvas Design

Inputs ★

- If you choose non-surveyed country (Timor Leste) in round 2: What input would you provide?
- If you use semantic or keyword matching, adjust for different jurisdiction and languages.
- Consider the format of your input.

Outputs ★

- Extract entire relevant section as text before and after the main identified sentence(s) can provide important context.
- What if multiple sections in the same document are relevant? (which they often are)

Evaluation ★

- What about new discovery: Did the tool extract a relevant text that human researcher missed? Aka does tool outperform humans.
- Top-N/ranking method: The list doesn't have to be ranked; all relevant laws identified by this method can move on to next stage. In the early stages of the tool development, it is advisable to review a broader set of results to reduce the risk of overlooking relevant legal instruments.
- RDTII database is as latest as 2025-early 2026 (SGP); you might discover new updates/regulations.

Risks

- High cost of tool is one of potential risk – especially if using LLM to process documents.
- Semantic difference: different jurisdictions might have different vocabulary
- Sector level vs horizontal: many said that extracting sector level notices are risk. They are not.
- Scalability?
- Sometimes law pdfs are no longer available on government site but the official document is hosted by third party website. What would you do then? + web archive
- Third party review: What if RDTII is the broadest and up to date dataset?
- Accuracy of OCR while reading scanned images of non-English documents?
- Misinterpretation / wrong-indicator mapping (highest): Need to be careful as in some countries the same legal text applies across several indicators. Not mutually exclusive.
- What would you do if there are several mentions of a new law in secondary sources, but the document is not available yet (blocked on website, not uploaded online?)

D • Hands on exercise 1: What is wrong with this output?

Indicator 11.1: Lack of transparent technical standards for India

Information Technology Act, 2000 — Section 70B

a) Instrument Citation

- Title: The Information Technology Act, 2000 (Act No. 21 of 2000), as amended
- Type: Act (Primary Legislation)
- Issuing Authority: Parliament of India; administered by MeitY
- Section / Article / Clause: Section 70B (establishment of CERT-In; functions; power to issue directions)
- Effective Date: In force from 17 October 2000 (date of enactment); section 70B inserted by the Information Technology (Amendment) Act, 2008 (Act No. 10 of 2009), effective 27 October 2009 per gazette notification
- In-force status: Confirmed in force as of 2026; available on MeitY domain ([meit.gov.in/static/uploads/2024/03/ITbill_2000.pdf](https://www.meit.gov.in/static/uploads/2024/03/ITbill_2000.pdf)). Section 66A (previously used for online speech) was struck down in *Shreya Singhal v. UOI* (2015) and is omitted — not relied upon here. Section 70B remains operative.
- Official URL: https://www.meit.gov.in/static/uploads/2024/03/ITbill_2000.pdf

b) Finding

Section 70B(1) of the IT Act, 2000 “provides that the Central Government shall appoint an agency of the Government to be called the Indian Computer Emergency Response Team,” with the Central Government having appointed CERT-In via gazette notification of 27 October 2009.

Section 70B(4) provides that CERT-In shall perform prescribed functions in cybersecurity, including: collection, analysis and dissemination of information on cyber incidents; alerts; emergency measures; coordination; and issuing guidelines, advisories, vulnerability notes and whitepapers. The Act empowers CERT-In to issue binding directions (Section 70B(6)) but does not establish a standard-setting governance framework with defined participation rules for stakeholders.

Hint: Look up the act and its contents.

Type your answer in the
zoom poll! You have 5
minutes

D • Hands on exercise 1:

What is wrong with this output?

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Answer: Section 70 B (1) and (4) does not appear in this act.

This is an example of hallucination/wrong retrieval

D · Hands on exercise 2:

What is wrong with this output?

Indicator 9.1 for Singapore: Blocking or filtering commercial web (exception of illegal activities)

- 1) Any blocking measure
- 2) Any filtering measure
- 3) No cases of blocking nor filtering (except internationally agreed illegal content/child pornography)

URL: <https://www.imda.gov.sg/regulations-and-licensing-listing/content-standards-and-classification/standards-and-classification/internet>

Title: Internet Regulatory Framework – IMDA website

Type: other

Relevance Score: 1.00

Excerpt 1

Location: Document **Text:** > “Unauthorised websites which promote, facilitate or advertise remote gambling are blocked under the Ministry of Home Affairs’ Gambling Control Act 2022. The regulation of remote gambling activities under the Act serves to maintain law and order and protect young persons and other vulnerable persons from being harmed or exploited by remote gambling.”

Relevance Notes

Official regulator site confirms that **unauthorised remote gambling websites** (a type of commercial online service) are **blocked by government directive** under a specific law (Gambling Control Act 2022). This shows a government-mandated mechanism to restrict access to certain commercial web services (online gambling).

Hint: is online gambling legal in Singapore?

**Type your answer in the
zoom poll! You have 5
minutes**

D · Hands on exercise 2:

What is wrong with this output?

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Relevance Notes

Official regulator site confirms that **unauthorised remote gambling websites** (a type of commercial online service) are **blocked by government directive** under a specific law (Gambling Control Act 2022). This shows a government-mandated mechanism to restrict access to certain commercial web services (online gambling).

Answer: Online gambling is already illegal in SGP. This will not be recorded as it is not considered to be commercial web content and doesn’t create any substantive burden on businesses engaged in digital trade.

Understanding country specific context is necessary.

However, it is not totally wrong, it's good output because it captured the fact that it is government blocked service provider.

D · Hands on exercise 3:

Which output would you record in the RDTII database?

**Goal of this exercise: To help you be able to evaluate whether new discovery is valid.
Can you think like a lawyer? Let's test it!**

Let's look at Indicator 2.1 for Malaysia:

2.1 Foreign exclusions from public procurement related to ICT goods and digital services

1: For any legislative measure excludes **foreign firms** from public procurement under any circumstances, or more than one measure under category (2)

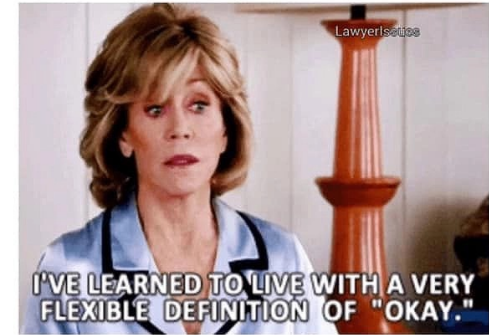
0.5: For any legislative measure excludes a specific (group) of **foreign firm(s)** from public procurement

0: No measure (when there is only a legal basis that grants the government the power to exclude foreign firms)

I will show you 2 output tables to compare and will ask you 2 questions.

"Are you okay?"

Lawyers:



E · Hands on exercise 3:

Which output would you record in the RDTII database?

Option A:

Examine the tables further: Please scan
QR for google sheet with both output
tables



Law / Regulation	Coverage	Analysis	Suggested Score
Treasury Circular PK1.1 (Perolehan Kerajaan – Dasar Bumiputera)	ICT goods, software, cloud services, IT consulting, digital services	PK1.1 – Quotation Procedures for Supplies and Services (RM50,000–RM100,000): Quotations are restricted to registered Bumiputera suppliers for procurements within the specified threshold. Foreign firms cannot generally qualify as Bumiputera suppliers and are therefore excluded from these procurement opportunities. Excludes a category of foreign firms from participating in certain public procurements, including ICT-related procurements falling within the threshold.	1
Treasury Circular PK1.5 (Syarikat Bumiputera Policy)	ICT goods and services, software procurement, digital transformation projects	PK1.5 – Qualification Requirements for Bumiputera Company Status: Firms seeking Bumiputera procurement status must satisfy Bumiputera ownership and management criteria; recent amendments reportedly require the Chief Financial Officer/Financial Controller to be Bumiputera. Restricts access to procurement opportunities reserved for Bumiputera firms. Foreign-owned firms generally cannot satisfy these requirements.	1
CPTPP Annex 15-A (Malaysia Schedule), Section G, Paragraph 2(a)–(b)	ICT goods and services procured by covered government entities	Section G, Paragraph 2(a)–(b): Malaysia reserves the right to maintain Bumiputera preferences and apply price preferences to Bumiputera suppliers and manufacturers in government procurement. Confirms the continued operation of procurement preferences that may disadvantage or exclude foreign suppliers from certain procurement opportunities.	1
Financial Procedure Act 1957	Government procurement generally, including ICT procurement	No provision identified that excludes foreign firms from public procurement. The Act governs financial administration, public funds, and public property management. Enabling legislation only; no evidence of foreign supplier exclusion.	0
Government Contracts Act 1949	Government contracts generally, including ICT and digital services contracts	Sections 2–5: Govern execution and authorization of government contracts. No provision identified restricting participation by foreign suppliers. Section 5 expressly contemplates contracts made outside Malaysia. No exclusion of foreign firms identified.	0

E · Hands on exercise 3:

Which output would you record in the RDTII database?

Option B:

Examine the tables further: Please
scan QR for google sheet with
both output tables



Law / Regulation	Coverage	Analysis	Suggested Score
Financial Procedure Act (Act 61) 1957	Horizontal	The Financial Procedure Act (Act 61) 1957, remains Malaysia's main legal instrument for financial matters including government procurement. For specific works, if local contractors do not have the expertise and capability, tenders may be called on a joint venture basis between local and foreign contractors to encourage the transfer of technology. International tenders for works may only be called when local contractors do not have the expertise and capability, and a joint venture is not possible. Section 6: Management and control of public finances, including the supervision of all matters related to the financial affairs of the Federation. This section provides overarching guidance for the use of public funds in procurement activities, including ICT goods and digital services. Section 13: No money shall be withdrawn from the Consolidated Fund without proper authority, which also implies that all procurement (including ICT-related services) would require authorization under public financial management rules. Section 15A: Controlling officers are responsible for ensuring that public procurement follows the prescribed rules and that expenditures are justified. Section 36: The regulations made under this Act may provide additional guidance on procurement rules, including preferences or exclusions for certain sectors like ICT.	1
Government Contracts Act (Act 120) 1949	Horizontal	The Government Contracts Act (Act 120) 1949 outlines the authority and procedures for making contracts on behalf of the Government of Malaysia, as well as state governments. According to Section 2, government contracts must be made in writing and signed by an authorized minister or public officer. This framework provides the legal structure for government procurement, including potentially any exclusions related to specific sectors such as ICT. Exemptions and provisions for handling contracts under state and federal government could allow specific exclusions based on strategic or national security concerns, especially in ICT and digital services procurement. Section 2: Contracts made on behalf of the Government of Malaysia must be in writing and signed by a Minister or a public officer authorized by a Minister. This covers all government contracts, including procurement. If specific exclusions for ICT goods and services are issued, they would be aligned with this section for signing and enforcement. Section 3: Similar provisions for State Government contracts, signed by the Chief Minister or a public officer duly authorized by the Chief Minister. Section 5: Contracts made outside Malaysia must also adhere to this structure when made on behalf of the Government.	0
Treasury Directive 1966	Horizontal	There are no explicit articles or sections in the Financial Procedure Act (Act 61) 1957, Government Contracts Act 1949, or other statutory laws provided, that specifically reference the requirement for international tenders only when local expertise is unavailable or the encouragement of joint ventures between local and foreign companies in sectors like ICT.	0

E · Hands on exercise 3: Which output would you record in the RDTII database?

**You have 5 minutes to review! The
voting opens afterward.**

Evaluation Checklist:

- Are all recorded documents domestic laws/regulations?
- Does the law truly show preference against foreign suppliers?
- Are the scores assigned to each entry, correct? Do they match the explanation?
- Should a document that has no regulatory power but has directive power be recorded?



Examine the tables further:
Please scan QR for google sheet
with both output tables

E • Hands on exercise 3:

Which output would you record in the RDTII database?

Which output would you record in your database?

- A
- B
- None



Cast your vote in the zoom poll!

Which output is AI?

- A
- B



Cast your vote in the zoom poll!

E · Hands on exercise 3:

Which output would you record in the RDTII database?

Output A is from LLM and Output B is from Human researcher

In output A, AI correctly identified financial procedure Act and Government contracts act but missed treasury directive.

What was wrong in Output A?

1. Preference for Bumiputera suppliers is not a preference solely against foreign suppliers. It applies equally to other domestic suppliers too.
2. CPTPP Annex 15-A (Malaysia Schedule), Section G, Paragraph 2(a)–(b) is not a domestic law
3. Financial Procedure Act 1957 scored at 0: should be 1 as it does state that *International tenders for works may only be called when local contractors do not have the expertise and capability, and a joint venture is not possible.*

Extra: Why treasury directive matters? Even if not scored, it must be recorded: Think back on the economy's legal system!

Government procurement in Malaysia is not governed by a single law but is primarily managed through Treasury Instructions and Treasury Circulars issued by the Ministry of Finance (MOF). Although these instruments guide procurement processes, they lack the force of regulatory law.

However, key legislation applies to aspects of procurement, including:

1. The Financial Procedure Act 1957 (FPA), which regulates public finance management under Section 6 and allows the creation of procurement-related regulations under Section 36
2. The Government Contracts Act 1949 (GCA), which governs the formation of contracts on behalf of the Government under Section

Source: <https://resourcehub.bakermckenzie.com/en/resources/public-procurement-world/public-procurement/malaysia/topics/1-the-laws>

How to prevent these errors?

1. Better prompt design
2. Mix LLM and ML methods.
3. Give the tool a comprehensive knowledge base
4. And more..



ESCAP
Economic and Social Commission
for Asia and the Pacific

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Thank you! Looking forward to seeing your tools!

For any queries reach me at nikita.shahu@un.org

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